

EXPERIMENT – 11

CODE:

```
#include <reg51.h>

sbit LED1 = P1^0;
sbit LED2 = P1^1;

void delay()
{
    unsigned int i, j;

    for(i = 0; i < 200; i++)
        for(j = 0; j < 1000; j++);
}

void main()
{
    while(1)
    {
        LED1 = 1;

        LED2 = 0;

        delay();

        LED1 = 0;

        LED2 = 1;

        delay();
    }
}
```

```
}  
  
}
```

OUTPUT:

The screenshot displays the uVision IDE interface for a project named "LAB-11". The main window shows the disassembly of a C program. The program includes a header file "reg51.h" and defines two functions: "delay()" and "main()". The "delay()" function is a simple loop that delays for a specified time. The "main()" function calls "delay()" and then returns.

The registers window shows the state of registers r0-r7 and system registers. The registers r0-r7 are all 0x00. The system registers are also 0x00. The command window shows the load command: "Load "E:\\SIMATS COURSES\\EMBEDDED SYSTEMS\\EMBEDDED C\\Objects\\LAB-11".

The call stack shows the current function is "DELAY" at location C:0x0800. The main function is also visible in the call stack at location C:0x081D.

The Parallel Port 1 window shows the port address 0xFD and the pins 0, 1, 2, 3, 4, 5, 6, 7. The pins 0, 1, 2, 3, 4, 5, 6, 7 are all checked.

E:\SIMATS COURSES\EMBEDDED SYSTEMS\EMBEDDED C\LAB-11\proj - uVision [Non-Commercial Use License]

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x03
r5	0xe8
r6	0x00
r7	0xc8
sys	
a	0x00
b	0x00
sp	0x09
sp_max	0x09
dptr	0x0000
PC	0x0800
states	3288422
sec	1.644211
psw	0x00

Disassembly

```
4: void delay()
5: {
6:   unsigned int i,j;
7:   for(i=0;i<200;i++)
8:   {
9:     void delay()
10:    {
11:      unsigned int i,j;
12:      for(i=0;i<200;i++)
13:      {
14:        for(j=0;j<1000;j++);
15:      }
16:    }
17:  }
18: }
19: void main()
20: {
21:   LED1 = 1;
22:   LED2 = 0;
23:   delay();
24:   LED1 = 0;
25:   LED2 = 1;
26:   delay();
27: }
```

STARTUP.AS1

```
1#include <reg51.h>
2sbit LED1 = P1^0;
3sbit LED2 = P1^1;
4void delay()
5{
6  unsigned int i,j;
7  for(i=0;i<200;i++)
8  {
9    for(j=0;j<1000;j++);
10 }
11 void main()
12 {
13   while(1)
14   {
15     LED1 = 1;
16     delay();
17     LED1 = 0;
18     LED2 = 1;
19     delay();
20   }
```

Parallel Port 1

Port 1

P1: 0xFE 7 Bits 0

Pins: 0xFE

Command

Running with Code Size Limit: 2K

Load "E:\SIMATS COURSES\EMBEDDED SYSTEMS\EMBEDDED C\Objects\LAB-11"

Call Stack - Locals

Name	Location/Value	Type
DELAY	C:0x0800	
MAIN	C:0x081D	

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE

Simulation

t1: 1.64421100 sec L7 C1 CAP NUM SCRL OVR R/W

1000 AM 7/31/2026